

CLINICAL ARTICLE

Gynecology

Impact of Beyond Fistula programming on economic, psychosocial and empowerment outcomes following female genital fistula repair: A retrospective study

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Abstract

Objective: To retrospectively assess changes in economic status, psychosocial status and empowerment among women who participated in Beyond Fistula reintegration programming following fistula repair.

Methods: We conducted a retrospective study among 100 Beyond Fistula program participants capturing sociodemographic characteristics, obstetric and fistula history, program participation, and our primary outcomes: economic status, psychosocial status, and empowerment via quantitative survey at two time points: before program participation and currently. Data were collected from November 2020 to July 2021 from 2013 to 2019 program participants. We compared outcomes across these two time points using paired *t* tests or McNemar's tests.

Results: The proportion of individuals owning property (28.0% vs. 38.0%, $P=0.006$), having a current source of income (19.0% vs. 56.0%, $P<0.001$), and saving or investing income (11.0% vs. 37.0%, $P<0.001$) increased significantly from pre- to post-programming. We also identified statistically significant increases from pre- to post-programming in self-esteem (5.0 [IQR 4.0–5.0] vs. 5.0 [IQR 5.0–5.0], $P<0.001$), reintegration (53.0 [IQR 43.0–69.0] vs. 65.0 [IQR 51.0–72.0], $P<0.001$) and level of input into household economic decision making (2.0 [SD 1.0] vs. 2.3 [SD 1.0], $P=0.004$).

Conclusion: Beyond Fistula programming likely improved economic status, psychosocial status, and empowerment of participants. Post-surgical interventions incorporating a holistic approach can advance recovery through supporting psychosocial and economic wellbeing and should be offered to women undergoing genital fistula repair.

KEYWORDS

economic empowerment, female genital fistula, Kenya, mental health, obstetric fistula, psychosocial support, reintegration, vocational training

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1 | INTRODUCTION

Female genital fistula is a preventable morbidity commonly caused by prolonged obstructed labor, iatrogenic, or traumatic etiologies in lower-resource settings, presenting as urinary or fecal incontinence. Measurement is challenging;^{1,2} global prevalence estimates range from 500 000 to 2 million, with annual incidence up to 100 000.^{3,4} Rates are highest where women's access to timely high-quality obstetric care is limited.⁵ Population-based surveys suggest 0.1%–2.0% of reproductive-aged women in sub-Saharan Africa have experienced female genital fistula,⁶ including 1% in Kenya.⁷

Women with genital fistula experience severe physical and psychosocial consequences, potentially including gynecological, urological, neurological, gastrointestinal, dermatological, musculoskeletal, and psychological features.⁸ They are often stigmatized and unable to participate in social and economic spheres.^{9–11} Loss of employment and income-generating opportunities results in significant economic consequences, increasing vulnerability and poverty.¹² Gender inequalities and vulnerable social status further structures risk.^{13,14} Focused recent efforts have significantly improved community awareness regarding fistula development and treatment access in Kenya, tripling annual surgeries to 1500 by 2017.^{15,16} Kenya has about 20 trained fistula surgeons providing specialized care in at least six hospitals countrywide.^{17,18}

Post-surgical reintegration programming that incorporates a holistic approach supporting women's optimal physical, social and psychological wellbeing may optimize recovery, particularly among women with persistent symptoms.^{4,19–21} An evidence gap around best practices for women's post-surgical reintegration programming exists. Thus, in the present study, we sought to retrospectively assess changes in economic and psychosocial status among women who underwent surgical genital fistula repair at the Gynocare Women's and Fistula Hospital in Eldoret, Kenya and subsequently participated in the Beyond Fistula reintegration program.

2 | METHODS

2.1 | Study setting: Gynocare and Beyond Fistula

Gynocare Women's and Fistula Hospital provides referral services for a ~10 million catchment population, with a particular focus on fistula surgery. It is a FIGO-certified fistula surgery training center where fistula surgery is freely provided through stakeholder donations (e.g., Dr. Hillary Mabeya, Fistula Foundation, Direct Relief, and Walfriede Hospital, Germany). Beyond Fistula (est. 2012) is a nonprofit organization providing counseling and educational/vocational training programs to women after fistula surgery to support their economic empowerment.^{22,23} During hospitalization, women receive three counseling sessions: to prepare for the surgery, to discuss reintegration, and to review home care. Counselors identify potential Beyond Fistula beneficiaries based on need, feasibility, and interest and maintain a list of potential beneficiaries. Beyond Fistula provided programming to approximately 300 women per year prior to the COVID-19 pandemic. Women enroll in the Beyond Fistula

program at varying periods post-surgery ranging from 2 months to 1.5 years or more, depending on program capacity and individual needs.

2.2 | Beyond Fistula intervention

The Beyond Fistula program consists of counseling combined with various locally developed livelihood training (Table 1).

2.3 | Research context

This evaluation was substantially affected by the COVID-19 pandemic and associated mitigation efforts. Kenyan government-imposed protocols to reduce COVID-19 decreased health-related harms but had adverse social and economic impacts.^{24–26} Disproportionate pandemic impacts on women included greater economic consequences, greater social disruptions (school, health, mental health), higher food insecurity, and five-fold increase in intimate partner violence.^{24,27,28}

2.4 | Study participants

We recruited 100 women who had undergone fistula surgery at Gynocare and participated in Beyond Fistula programming between 2013–2019. Data were collected from November 2020–July 2021. Sample size prioritized feasibility and is adequate to detect a 25% minimum difference in pre-/post outcomes at $\alpha=0.05$ and $1-\beta=0.8$, using McNemar's test.

2.5 | Study procedures

Participant recruitment occurred through simple randomization of all prior Beyond Fistula participant names until target sample size achievement. Potential participants were contacted via phone by research assistants to describe the study objectives, risk and benefits, and to invite them to participate. None invited refused. After agreeing to participate, an in-person meeting was scheduled at a mutually convenient location where privacy could be ensured for survey administration. One interviewer from our team of 10 former Beyond Fistula participants trained in collecting sensitive data met with the research participant, led them through the informed consent process, obtained written confirmation of consent, and verbally administered the survey. Study data were captured on paper forms and entered into secure REDCap electronic data capture system.²⁹ Participants were provided with beans and maize (~\$5 value) in appreciation.

2.6 | Study measures

Primary outcomes were economic status, psychosocial status and empowerment. Economic status was measured through asset ownership

TABLE 1 Beyond Fistula intervention components.

Program	Rationale	Components
Psychological support		
Counseling	Women with fistula are stigmatized. The social traumas of fistula cause intense psychological wounds that require counseling to heal. Beyond Fistula provides one-on-one therapy and group therapy to address this trauma. Counseling is provided per individual need while participating in livelihood training.	Daily individual counseling sessions (20–40 min) Daily group counseling (30 min; 5–10 women) Free room and board in Beyond Fistula house
Livelihood training		
Vocational training	Beyond Fistula provides vocational training in tailoring, hairdressing, and basket weaving. Participants receive a business grant upon completion, fundraised in partnership with Benevolent. Many of the vocational graduates go on to full-time positions in their fields, or venture into business ownership by participating in further business training.	Tailoring or hairdressing training: 1 year funded study at a local vocational college Catering program: 2 years funded study at a local vocational college Free room and board in Beyond Fistula house, if needed A business grant upon completion (amount varies)
Business training	Beyond Fistula provides business training for women who have prior work experience in a specific field and seek to advance their careers by starting their own small businesses.	3 day small-business ownership module Small grant from Beyond Fistula upon completion, ~\$50 USD
Farm skills program	Farming is a common means of earning income in rural Kenya. Many fistula survivors have a background in farming and, once healed, are eager to turn their experience into an income-generating business.	3 day farm skills training seminar taught by our partner, Farming God's Way Small grant from Beyond Fistula upon completion, ~\$50 USD
Scholarship program	Where fistula survivors dropped out of school (e.g., at pregnancy or fistula development), Beyond Fistula provides an educational scholarship in partnership with The School Fund which pays for uniforms, school fees, and books.	Pays for uniforms, school fees, and books Free room and board in Beyond Fistula house if needed

(i.e., land, animals, machinery, and technology), employment status (e.g., formal employment for wages, informal employment for wages, self-employed, housewife, not employed) and income. Psychosocial status included assessment of self-esteem (range 1–5), reintegration (range 0–100),³⁰ and depressive symptom frequency (range 1–4). Empowerment was assessed by participation in household decision making (domains: use of money participant earned, healthcare for participant, major household purchases, visit to family or relatives and input into household food crop farming, cash crop farming, livestock farming, non-farm economic activities, and wage and salary employment) and intimate partner violence.³¹ Outcomes referred to two time points: prior to Beyond Fistula participation (post-surgery) and currently. Covariates included sociodemographic demographics (i.e., age, relationship status, educational attainment), obstetric and fistula history (parity, pregnancy outcomes, number of surgeries, surgical outcomes, physical symptoms), and perspectives on Beyond Fistula programming. Administrative data supplemented self-reported participation data.

2.7 | Statistical analysis

We described the sociodemographic characteristics and fistula history of the study population, presenting mean and standard deviation or median and interquartile range, as appropriate, for continuous variables and percentages for categorical variables. We conducted bivariable analyses comparing outcomes by covariates of interest pre- to

post-participation using paired *t* tests for continuous variables (e.g., mean household decision making score, average monthly income, etc.) and McNemar's test for categorical variables (e.g., employment status, asset ownership, and so on). Fisher exact tests were used to compare program impact perspectives by incontinence status. Analyses were conducted in Stata version 16 (College Station, TX, USA), and differences where $P < 0.05$ was considered statistically significant.

2.8 | Ethics review

The study protocol and all study documents were reviewed and approved by the University of California San Francisco's Human Research Protection Program (19-29502) and the Moi Teaching and Referral Hospital/Moi University Institutional Research and Ethics Committee (IREC/2020/000139). Written confirmation of informed consent was obtained from all study participants.

3 | RESULTS

3.1 | Participant characteristics

Mean age was 38.5 years (SD 20.4) at study participation and 25.7 years (SD 10.9) at fistula development (Table 2). Most participants were married or in domestic partnership ($n = 65$; 65%),

TABLE 2 Participant sociodemographic characteristics, fistula history, and Beyond Fistula program participation, retrospective survey, Beyond Fistula program participants.

	N	%
Age ^a	38.5 (10.4)	
Age at fistula development	25.7 (10.9)	
Relationship status		
Single/never married	15	15.0
Married or domestic partnership	65	65.0
Widowed	9	9.0
Divorced/separated	11	11.0
Educational attainment		
None	11	11.0
Some primary	42	42.0
Completed primary	25	25.0
Some secondary	12	12.0
Completed secondary or higher	10	10.0
Past year employment		
Not employed ^b	46	47.0
Formal employment for wages	8	8.0
Informal employment for wages	1	1.0
Self-employed	45	44.0
Job	(n=54)	
Farming	9	16.7
Business	34	63.0
Other	11	20.3
Food insecurity level		
None	12	13.0
Marginal	6	6.0
Severe	82	81.0
Level of recovery from fistula		
Completely	71	71.0
Almost	5	5.0
Partially	16	16.0
Not at all	8	8.0
Current urine leakage	34	34.0
Current fecal leakage	25	25.0
More than one surgery ^a	39	39.0
Current fistula-related symptoms		
Pain during urination	23	23.0
Pain during sexual intercourse	22	22.0
Other vaginal or pelvic pain	25	25.0
Weakness	32	32.0
Difficulty walking	32	32.0
Skin irritation	22	22.0
Other	13	13.0
Beyond Fistula program		
Vocational training (sewing, hairdressing, catering)	6	6.0

TABLE 2 (Continued)

	N	%
Business training	45	45.0
Farm skills program	41	41.0
Both Business and Farm skills program	8	8.0
Year participated in Beyond Fistula		
2013–2016 ^c	5	5.0
2017	17	17.0
2018	46	46.0
2019	32	32.0
Time from surgery to program	(n=95)	
<1 year	36	37.9
1–2 years	33	34.7
3–5 years	12	12.6
More than 5 years	14	14.8

^aMean (SD).

^b16 of 43 not employed were housewives.

^c1 participant in 2013 and 4 participants in 2016.

and almost half had completed primary education ($n=47$; 47%). Participants were largely not employed ($n=46$; 46%) or self-employed ($n=45$; 45%) in the past year. Food insecurity was severe for 81% ($n=82$) of participants. Participant residence is displayed in [Figure 1](#).

Most participants considered themselves completely recovered from fistula ($n=71$; 71%). Current fistula-related symptoms reported included urine leakage ($n=34$; 34%), weakness ($n=32$; 32%), difficulty walking ($n=32$; 32%), fecal leakage ($n=25$; 25%), vaginal or pelvic pain ($n=25$; 25%), pain during urination ($n=23$; 23%), pain during sexual intercourse ($n=22$; 22%), and skin irritation ($n=22$; 22%). Participants completed Beyond Fistula's business training ($n=45$; 45%) or farm skills program ($n=41$; 41%). Time from surgery to program was <1 year ($n=36$; 38%), 1–2 years ($n=33$; 35%), and three or more years ($n=26$; 27%).

3.2 | Change in economic status across Beyond Fistula program participation

Comparisons between pre- and post-program indicators of economic status are presented in [Table 3](#). Statistically significant increases from pre- to post-program were observed in the proportion of participants reporting owning land (28.0% [$n=28$] vs. 38.0% [$n=38$], $P=0.006$) or animals (50.0% [$n=50$] vs. 60.0% [$n=60$], $P=0.031$), having a current job or source of income (19.0% [$n=19$] vs. 56.0% [$n=56$], $P<0.001$), and saving or investing income (11.0% [$n=11$] vs. 37.0% [$n=37$], $P<0.001$). Median income increased significantly from 0 (IQR 0–0) to 500 Kenyan Shillings (KSh; \$4.40 USD) (IQR 0–2700 KSh; 0–\$23.76 USD), with median difference of 0 (IQR 0–2000 KSh; 0–\$17.60 USD, $P<0.001$). No significant change was observed in the proportion of participants owning machinery or technology.

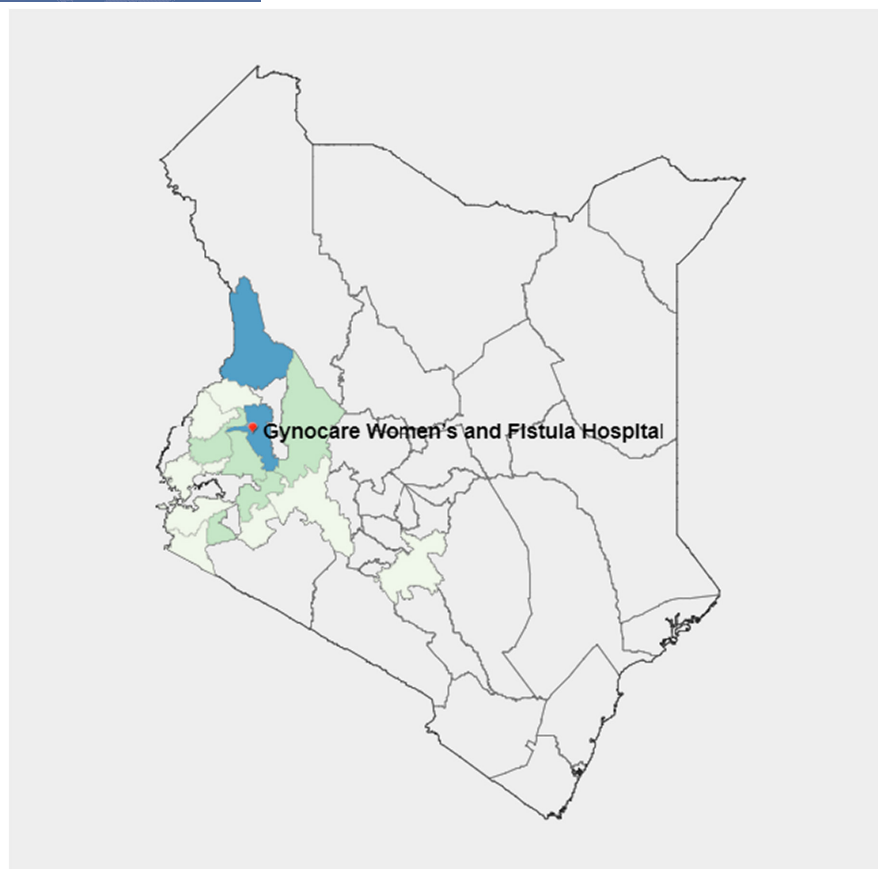


FIGURE 1 Participant residence, retrospective survey, Beyond Fistula program participants. Source: Program data and UN-OCHA Humanitarian Data Exchange Project.

3.3 | Change in psychosocial status across Beyond Fistula program participation

Pre- and post-program indicators of psychosocial status are presented in Table 4. Statistically significant increases from pre- to post-program were observed in median self-esteem (5.0 [IQR 4.0–5.0] vs. 5.0 [IQR 5.0–5.0], $P < 0.001$) and reintegration (53.0 [IQR 43.0–69.0] vs. 65.0 [IQR 51.0–72.0], $P < 0.001$). Prior to program participation, 65.7% ($n = 65$) of participants selected the highest option on the self-esteem scale, “I loved myself”, which increased to 85% ($n = 85$) post-participation ($P < 0.001$, not shown). No change was found in frequency of depressive symptoms.

3.4 | Change in empowerment across Beyond Fistula program participation

Pre- and post-program indicators of empowerment are presented in Table 4. Statistically significant increases from pre- to post-program were found in level of input into household economic decision making (2.0 [SD 1.0] vs. 2.3 [SD 1.0], $P = 0.004$) and women's involvement in decision making about how money they earned would be used (77% [$n = 76$] vs. 87% [$n = 87$], $P = 0.002$), decision making regarding major household purchases (74% [$n = 73$] vs. 82%

[$n = 82$], $P = 0.013$), and in decision making on visits to their family/relatives (79% [$n = 78$] vs. 86% [$n = 86$], $P = 0.003$). Observed reductions in intimate partner violence did not meet statistical significance. Prevalence of intimate partner violence varied substantially by current relationship status, with separated/divorced women reporting higher rates than those married or in partnership (Table S2).

3.5 | Program perspectives

Participants reported high acceptability of Beyond Fistula programming; 77% ($n = 77$) were very likely to recommend the program (Table 5). Perceived program impact was good, with most participants reporting a large or medium impact on financial status (45% [$n = 45$] and 29% [$n = 29$], respectively), social status (52% [$n = 52$] and 25% [$n = 25$]), emotional status (52% [$n = 52$] and 27% [$n = 27$]), and status in household (45% [$n = 45$] and 33% [$n = 33$]). About one-fifth of respondents indicated no impact across these domains; this was significantly higher for individuals reporting persistent incontinence (Table S1). Most participants reported that applying the knowledge and skills that they had gained from the program was easy (58% [$n = 58$]) or somewhat easy (17% [$n = 17$]). Challenges in implementing knowledge and skills included: costs

TABLE 3 Pre- and post economic indicators, retrospective survey, Beyond Fistula program participants.

	Pre-program		Post-program		P value
	n	%	n	%	
Own land	28	28.0	38	38.0	0.006
Own animals	50	50.0	60	60.0	0.031
Own machinery	6	6.0	6	6.0	1.000
Own technology	73	73.0	77	77.0	0.455
Current job/source of income	19	19.0	56	56.0	<0.001
Current save or invest income	11	11.0	37	37.0	<0.001
Average monthly income ^a					
Median (IQR)	0 (0-0)		500 (0-2700)		<0.001
Range	0-35 000		0-35 000		

^aIncluding individuals reporting no income. Exchange rate Kenyan Shillings 113.65 KSh:1 Dollar, this post-program median income \$4.40 US (IQR 0-\$23.76) and range for both time points 0-\$307.96 (note only two individuals reported present income above 9000 Ksh/\$79.19 US).

($n=7$), need for refresher training ($n=5$), environment ($n=2$), and other responsibilities ($n=4$). Recommendations to improve Beyond Fistula programming ($n=88$) included: none ($n=19$), continue current programming ($n=19$), gratitude for Beyond Fistula programming ($n=19$), increase financial support ($n=13$), increase community awareness ($n=12$), provide more skills training ($n=11$), and follow-up ($n=5$).

4 | DISCUSSION

Our findings suggest that participation in Beyond Fistula's psychological support and livelihood training programming is highly acceptable and positively impacted the economic and psychosocial status of participants. We observed statistically significant differences from pre- to post-program in ownership of land and animals, employment and income, savings, self-esteem, and involvement in varied domains of household decision making. Given fistula's broad consequences including negative impact on psychological health, social status and economic stability, holistic approaches to fistula treatment are recommended, including supporting psychosocial and economic recovery.^{4,20} Our study extends the limited evidence quantifying the potential impact of fistula reintegration programming.

Comparative literature is sparse, particularly robust study designs;²⁰ however, findings from other psychosocial support and economic empowerment interventions for fistula are consistent with our study, reporting improved self-esteem, functioning, ability to work, and mental health. For example, individual and group counseling is generally associated with reduced depression and suicidal ideation and increased self-esteem as reported studies in Eritrea, South Sudan and Tanzania, although not all comparisons were statistically significant.³²⁻³⁴ Kenya's Country-Wide Fistula Treatment Network found that a psychosocial counseling, social support, and economic empowerment intervention improved socializing ability, ability to work and normal functioning with a reintegration strategy including follow-up visits, community support groups, income

generating activities, vocational training, and educational support in the 12 months following repair.¹⁶ The significant increases in reintegration we identified across program participation are promising but contextualizing these findings by assessment time is important. For example, observational research has identified larger reintegration increases immediately following surgery.³⁵⁻³⁷ A Ugandan study reported mean reintegration of 33.2 (SD 20.4) at surgery and 70.4 (SD 27.1) at three-months post-surgery compared to our pre- to post-participation median increase from 53 (IQR 43-69) to 65 (IQR 51-72). This may be explained by different assessment points or different levels of participant socioeconomic status among participants in this study, as one-third of both study populations had residual incontinence. Further research to characterize reintegration program-attributable outcomes will need to incorporate contextual and individual factors through more robust study designs.

Improvements in economic status were largest in employment and savings or investment of income whereas increases in asset ownership and actual income were smaller. This is not surprising given the time since participation in Beyond Fistula programming was relatively short for most participants, the fact that most individuals participated in the two shorter trainings (business and farm skills), and their rural contexts, combined with the economic impact of the COVID pandemic and mitigation strategies. Findings from poverty reduction programming in sub-Saharan Africa show that gender plays a large role in structuring women's opportunities for labor market participation through both industry engagement options and navigating gendered unpaid household work structures which are largely the burden of women and responsible for significant opportunity cost.³⁸ Future research on reintegration programming impact should employ a gender lens and incorporate nuanced data on program participation, post-participation support, post-programming activity engagement, as well as attend to general local market opportunities.

Our findings on women's involvement in household decision making were similar or higher than regional estimates from the 2014 Kenyan Demographic and Health Survey (DHS). For example, the

TABLE 4 Pre- and post social indicators, retrospective survey, Beyond Fistula program participants.

	Pre-program		Post-program		P value
	n	%	n	%	
Self-esteem (median, IQR)	5.0 (4.0–5.0)		5.0 (5.0–5.0)		<0.001
Reintegration (median, IQR)	53.0 (43.0–69.0)		65.0 (51.0–72.0)		<0.001
Depressive symptoms (n = 99)					
Very or somewhat often	61	61.6	62	62.0	0.808
Very or somewhat seldom	38	38.4	38	38.0	
Level of input into household economic decision making (mean, SD)	2.0 (1.0)		2.3 (1.0)		<0.001
Household decision making by domain					
How money respondent earned would be used (n = 99)					0.002 ^b
Respondent	45	45.5	55	55.0	
Husband/partner	13	13.1	10	10.0	
Respondent and husband/partner jointly	31	31.3	32	32.0	
Someone else	10	10.1	3	3.0	
Decisions about healthcare for respondent					0.090 ^b
Respondent	29	29.3	29	29.0	
Husband/partner	16	16.2	15	15.0	
Respondent and husband/partner jointly	34	34.3	41	41.0	
Someone else	20	20.2	15	15.0	
Major household purchases					0.013 ^b
Respondent	41	41.4	43	43.0	
Husband/partner	15	15.2	11	11.0	
Respondent and husband/partner jointly	32	32.3	39	39.0	
Someone else	11	11.1	7	7.0	
Visit to respondent's family/relatives					0.003 ^b

TABLE 4 (Continued)

	Pre-program		Post-program		P value
	n	%	n	%	
Respondent	37	37.4	39	39.0	
Husband/partner	12	12.1	7	7.0	
Respondent and husband/partner jointly	41	41.4	47	47.0	
Someone else	9	9.1	7	7.0	
Intimate partner violence (n = 84)			(n = 81)		
Said/done something to humiliate/insult ^a	15	17.9	9	11.1	0.058
Threaten to hurt/harm you or someone close to you	13	15.5	11	13.6	0.317
Harmed you physically	11	13.1	9	11.1	0.157
Prohibited you from getting a job, going to work, trading or earning money	11	13.1	7	8.6	0.103
Take your earnings or property against your will	7	8.3	5	6.2	0.317
Throw you out of the house	8	9.5	6	7.4	0.317
Forced to do sexual things when did not want to	11	13.1	13	16.0	0.414

^aIPV questions were responded to by those with a partner at the relevant time reference.

^bP value for woman's involvement in decision making (decision made by respondent or respondent and husband/partner jointly vs. husband/partner alone or someone else).

DHS reported 58.0% of women decided alone how their earnings would be spent (vs. 55.0% in our study) and 31.5% decided jointly with their husbands (vs. 32.0%).¹⁴ Involvement in own healthcare decision making was slightly lower than the 70.3% regional estimate pre-participation (63.6%) but similar post-participation (70.0%). Decision making involvement in major household purchase (73.7% and 82% pre-/post vs. 64.8%) and visits to the woman's family/relatives (78.8% and 86.0% pre-/post vs. 62.5%) were higher in our study. Intimate partner violence was generally lower than regional or national estimates during both assessment points; this may be because most (65%) of our study participants were married or in partnership or it may be underreported given the sensitivity of the topic.

TABLE 5 Program acceptability and reported impact, retrospective survey, Beyond Fistula program participants.

	N	%
Likelihood of recommending Beyond Fistula programming		
Very Likely	77	77.0
Likely	22	22.0
Unlikely	1	1.0
Program impact on financial status		
None	21	21.0
A small impact	5	5.0
A medium impact	29	29.0
A large impact	45	45.0
Program impact on social status		
None	20	20.0
A small impact	3	3.0
A medium impact	25	25.0
A large impact	52	52.0
Program impact on emotional status		
None	18	18.0
A small impact	3	3.0
A medium impact	27	27.0
A large impact	52	52.0
Program impact on status in household		
None	18	18.0
A small impact	4	4.0
A medium impact	33	33.0
A large impact	45	45.0
Ease of applying knowledge/skills gained from program		
Difficult	3	3.0
Somewhat difficult	10	10.0
Neutral	12	12.0
Somewhat easy	17	17.0
Easy	58	58.0
Challenges experienced in implementing knowledge/skills		
No	83	83.0
Yes	17	17.0

While most participants implemented the skills they learned from Beyond Fistula, one-fifth reported no change in impact. Ongoing challenges for individuals with continued incontinence are consistent with one Ugandan study which reported significantly lower reintegration score compared to those with symptoms resolution,¹⁹ yet another found significant improvement in emotional distress, quality of life and health satisfaction among women with incurable fistula participating in TERREWODE reintegration programming.³⁹ Reduced program impact among women with ongoing incontinence emphasizes the critical need to optimize fistula care to increase positive outcome likelihood, as surgical success rates decrease with subsequent surgery,⁴⁰ and the importance of this small but high-need subgroup.

Our study extends the limited evidence base on the impact of reintegration programming following genital fistula repair in lower-resource settings; however, interpretation of findings must acknowledge our limitations. First, our quasi-experimental design lacked a control group and participants retrospectively assessed pre-program status, increasing potential recall bias. Our study was bolstered by use of objective quantitative assessment instead of participant self-reported change. We sought to reduce social desirability bias likelihood and increase participant comfort through employing interviewers from the target population; however, interviewers may have been perceived to be affiliated with the program. The COVID-19 pandemic context and related logistical challenges limited inclusion to only participants reachable by phone, and current economic and psychosocial status reported by participants may also reflect the adverse social and economic impacts of the pandemic which disproportionately affected women.^{24,27,28} This suggests conservative pre- to post-participation changes.

5 | CONCLUSIONS

Our findings suggest that Beyond Fistula program participation improved economic status, psychosocial status, and empowerment. Holistic interventions to overcome the negative physical, psychosocial, and economic impact of genital fistula such as Beyond Fistula are important for improving health and well-being of this population. Further research priorities include robust assessment of reintegration programming to inform program optimization.

AUTHOR CONTRIBUTIONS

Debra Matityahu and Hillary Mabeya were involved in Beyond Fistula program design and implementation. Alison M. El Ayadi and Debra Matityahu conceived of the study. Alison M. El Ayadi drafted the protocol and led development of the study design, supported by Debra Matityahu and Hillary Mabeya. Alison M. El Ayadi, Jessica Alway, Debra Matityahu, Susan Wilson, and Hillary Mabeya assisted in instrument development. Jessica Alway led data collection and field team supervision, helped by Alison M. El Ayadi, Debra Matityahu, Susan Wilson, and Hillary Mabeya. Celine Kichwen led the local research team. Alison M. El Ayadi managed data analysis, supported by Jessica Alway, Debra Matityahu, and Susan Wilson. Alison M. El Ayadi and Jessica Alway conducted literature review and drafted manuscript. All authors revised the manuscript and approved the submitted version.

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CONFLICT OF INTEREST STATEMENT

Debra Matityahu is the co-founder and executive director of Beyond Fistula, a 501(c) (3) nonprofit organization. Susan Wilson is the assistant director of project management at Beyond Fistula. Dr Hillary Mabeya is a surgeon at Gynocare Women's and Fistula Hospital, which provides surgical intervention for fistula repair. Authors report no other conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets generated during the current study are available from the corresponding author on reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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